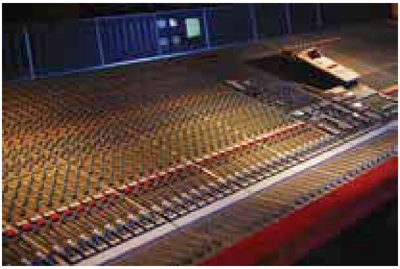
A typical home speaker consists of much more than public announcement speakers shown above.

A loudspeaker essentially converts an electric audio signal into a corresponding sound. The most widely used type of loudspeaker is the dynamic speaker. When an alternating current electric audio signal is applied to a coil of wire suspended in a circular gap between the poles of a permanent magnet, the coil has to move back and forth due to Faraday's law of magnetic induction, which causes an attached diaphragm to move and create sound waves by pushing the air. Where sound of high fidelity is needed, more than one loudspeaker transducers are often attached in the same enclosure, with each being responsible for a specific range of human audible frequency. In such builds, the individual speakers are called 'drivers' and the entire unit is called a loudspeaker.

Mixing console

Not many of us have seen the insides of a recording studio, but those who have would tell you about a massive console with almost countless keys and knobs. That's the digital audio mixing console used for combining, routing, volume adjustments, timbre adjustment and dynamics of many audio signals that are widely different.



A mixing console at the Cutting Room Recording Studios, NYC

These mixing consoles can be found at many locations being used for various purposes, like recording studios, PA Systems, sound reinforcement systems, nightclubs, broadcasting, television and movie post-production. An average mixer can combine the inputs from two microphones into a single amplifier, which in